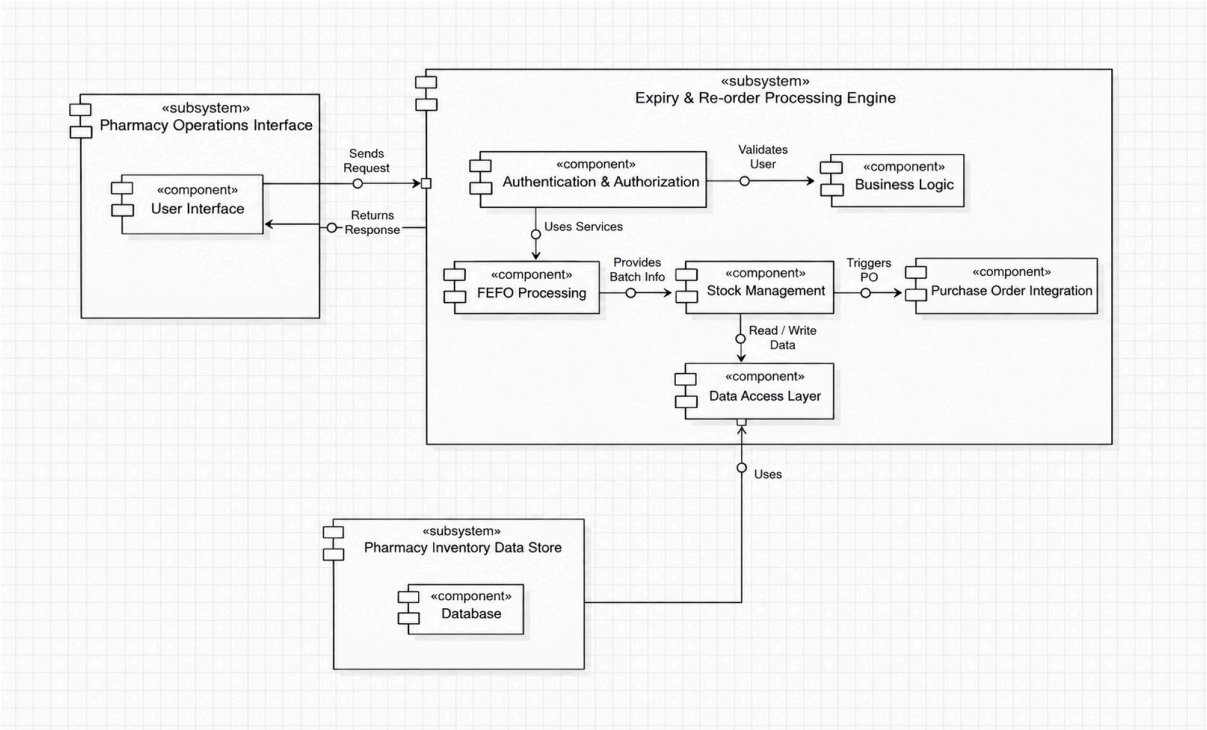
SE | LAB - 03

Pharmacy Expiry & Re-order Dispatch Engine

NAME: SHAIKH ABDULLAH MAHDI
SRN: PES2UG24CS452
SECTION: H



Architecture Selection: Layered Architecture

The system follows a Layered Architecture, specifically a 3 tier architecture with an additional integration/service layer. The components in your diagram naturally separate into presentation, business, and data responsibilities.

1. Presentation Layer

Component:

- User Interface

Responsibility:
Provides the interface through which pharmacy staff interact with the system and send requests/receive responses.

2. Business Logic Layer

Components:

- Authentication & Authorization
- FEFO Processing
- Stock Management
- Business Logic

Responsibility:
Handles user validation, expiry processing, inventory operations, and the main decision-making rules of the system.

3. Integration Layer

Component:

- Purchase Order Integration

Responsibility:
Connects the system with the purchase-order process and triggers purchase orders when required.

4. Data Access Layer

Components:

- Data Access Layer
- Database

Responsibility:
Handles reading and writing inventory information and provides controlled access to the pharmacy database.